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An integrated CatBoost–Transformer framework for multi-parameter multi-wavelength hourly aerosol retrieval from GOES-R observations

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ABSTRACT

The operational GOES-R aerosol product provides only single-band aerosol optical depth (AOD) and exhibits substantial diurnal biases, limiting high-temporal-resolution characterization of aerosol properties. Here, we develop an integrated CatBoost–Transformer (AeroCatTrans) framework for multi-parameter, multi-wavelength retrieval of aerosol optical properties from hourly geostationary observations. CatBoost ingests dynamic multi-source observations to handle heterogeneous inputs and multi-output regression, while the Transformer uses self-attention to refine these estimates by modeling complex inter-parameter relationships, enabling simultaneous retrieval of AOD (440–1020 nm), fine- and coarse-mode AOD (500 nm), single-scattering albedo (SSA), and asymmetry factor (AF) across 440–1020 nm. The framework achieves high and stable accuracy for hourly retrievals, with sample-based cross-validation correlation coefficients (CV-R) of 0.90–0.92 for multi-band AOD. Similarly strong performance is obtained for fAOD and cAOD (CV-$R = 0.93$ and 0.89, respectively), while moderate-to-high accuracy is achieved for SSA (0.53–0.72) and AF (0.88–0.93). The model also demonstrates generalization capability across different aerosol properties under spatiotemporal-blocked CV, with CV-R values of 0.75–0.81 (AOD), 0.83 (fAOD), 0.79 (cAOD), 0.32–0.56 (SSA), and 0.81–0.88 (AF). Leveraging these multi-parameter, multi-wavelength retrievals, we reveal pronounced diurnal variability in aerosol loading (AOD: 39%–59%), contrasted with much weaker variability in aerosol absorption and scattering anisotropy (SSA and AF: 4%–7%), indicating the relative stability of aerosol microphysical properties throughout the day. Furthermore, derived diagnostic indices enable robust identification of aerosol type and emission sources, enhancing process-level tracking of smoke evolution during extreme wildfire events. These results establish a new paradigm for continuous, high-resolution monitoring of aerosol properties and their diurnal dynamics from geostationary satellites, with important implications for air quality and climate studies.

1. Introduction

Atmospheric aerosols are suspensions of solid and liquid particles in the atmosphere that influence the Earth's radiative balance and play an important role in air quality, ecosystems, and human health (Chang et al., 2025; Schmale et al., 2021; Zhao et al., 2024; Zhong et al., 2024). Aerosol optical depth (AOD) is a widely used indicator of column aerosol loading in climate studies (Cherian and Quaas, 2020; Kang et al., 2015), and is extensively applied to constrain surface particulate matter (Chen et al., 2018b; Fan et al., 2024; Wei et al., 2023) and access aerosol radiative effects (Chang et al., 2025; Thorsen et al., 2020; Zhao et al., 2024). In addition, other aerosol optical properties, such as fine-

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mode AOD (fAOD), Ångström Exponent (AE), single-scattering albedo (SSA), and the asymmetry factor (AF), provide critical information on aerosol size, scattering, absorption, and composition, enabling improved quantification of aerosol-radiation interactions and environmental impacts (Dong et al., 2023; Yan et al., 2021).

Satellite remote sensing has achieved major advances in aerosol optical depth (AOD) retrieval, leading to mature algorithms such as dark target (DT), deep blue (DB), multi-angle implementation of atmospheric correction (MAIAC). These algorithms have been widely applied to polar-orbiting satellite platforms (e.g., MODIS and VIIRS), providing long-term and global AOD products (primarily at 550 nm) extensively used in climate and air quality studies (Hsu et al., 2013; Levy et al., 2015; Lyapustin et al., 2018; Yang et al., 2024). In contrast, satellite retrievals of other aerosol optical properties remain limited. Operational products beyond AOD are still scarce; for example, MODIS provides only fine-mode fraction (FMF) and AF over the ocean, while other parameters are available only at limited wavelengths (Levy et al., 2010; Luo et al., 2024). For instance, AE is typically available in the 550–860 nm wavelength range, whereas SSA is available only at 412, 480, and 660 nm. To address these limitations, recent studies have explored advanced retrieval approaches, such as the enhanced lookup table (LUT) methods (Cheng et al., 2012; Wang et al., 2015; Yan et al., 2021; Zhang et al., 2021), Generalized Retrieval of Aerosol and Surface Properties (GRASP) algorithm (Chen et al., 2018a; Dubovik et al., 2011; Dubovik et al., 2014), and machine- and deep-learning methods (Dong et al., 2023; Fan et al., 2025a, 2025b; Wang et al., 2025b). These approaches have enabled the retrieval of multiple aerosol parameters with expanded spatial coverage and improved accuracy. However, these studies mainly rely on polar-orbiting satellites with limited observation frequency (typically 1–2 times per day), which constrains their capability to capture diurnal aerosol variations and track rapid changes during extreme events.

Geostationary satellites offer a unique opportunity to overcome these temporal limitations by providing high-frequency full-disk observations at 10–15 min intervals. However, compared with polar-orbiting platforms, the development of aerosol retrievals from geostationary satellites remains relatively limited. Existing studies have primarily focused on AOD retrievals (mainly at 500–550 nm) from sensors such as the Advanced Baseline Imager (ABI) onboard the Geostationary Operational Environmental Satellite (GOES) and the Advanced Himawari Imager (AHI) onboard Himawari-8, using algorithms adapted from polar-orbiting approaches or based on simplified radiative transfer assumptions (Yoshida et al., 2018; Zhang et al., 2020). These AOD products often exhibit larger uncertainties than their polar-orbiting counterparts, particularly over land, due to challenges associated with surface reflectance, viewing geometry, and reduced spectral information (Wei et al., 2019; Yang et al., 2020; Zhang et al., 2020). Retrievals of aerosol optical properties beyond AOD from geostationary observations remain even more scarce (Jiang et al., 2024; Lee et al., 2025; Li et al., 2025; Yoshida et al., 2018).

Several representative studies have leveraged Transformer-based models for aerosol retrieval, demonstrating the growing potential of Transformer architectures for retrieving various aerosol optical and microphysical properties from satellite and lidar observations (Fan et al., 2025a, 2025b; Jiao et al., 2026; She et al., 2024, 2026; Zou and Müller, 2025). Although previous studies have developed retrieval algorithms for individual aerosol optical properties from geostationary satellites, comprehensive retrievals of multiple aerosol optical properties across multiple wavelengths at high temporal resolution remain limited. To address this limitation, we develop a novel multi-parameter, multi-wavelength aerosol retrieval framework for geostationary satellites based on a deeply integrated multi-task model that combines CatBoost and a Transformer (AeroCatTrans). Specifically, CatBoost is used to extract high-dimensional nonlinear features from GOES-16/ABI observations, which are subsequently encoded as sequential tokens and processed by a Transformer to model contextual information and long-range dependencies. The final retrievals are generated by two weighted branches, improving both overall accuracy and robustness. The proposed new framework enables the simultaneous retrieval of key hourly aerosol optical properties from GEOS-R observations over the Americas, including AOD at 440, 500, 675, 870, and 1020 nm; fine-mode and coarse AOD at 500 nm; and SSA and AF at 440, 675, 870, and 1020 nm. The resulting high-frequency, multispectral dataset enables unprecedented analysis of the diurnal evolution of aerosol optical properties. More importantly, it significantly enhances the characterization of diverse pollution events by capturing their distinct optical signatures, such as the strong absorption of wildfire smoke.

2. Materials

2.1. GOES-16/ABI observations and aerosol products

GOES-16, the first satellite in the GOES-R series, was launched on 19 November 2016, and became operational as the GOES-East observatory in December 2017 at 75.2°W, providing coverage from 135°W to 15°W. As a cornerstone of hemispheric environmental monitoring, its primary instrument, the ABI, is a 16-channel multispectral sensor spanning the visible, near-infrared, and infrared wavelengths, designed for continuous imaging of the Western Hemisphere. It provides a consistent observational basis for atmospheric studies, including aerosol monitoring, with full-disk coverage every 10 min and spatial resolutions ranging from 0.5 to 2 km (Table S1).

In this study, we utilized the GOES-16/ABI full-disk top-of-atmosphere (TOA) data, with a nominal spatial resolution of 2 km at nadir. To ensure high-quality inputs for aerosol retrieval, we applied the Satellite Application Facility on Nowcasting and Very Short Range Forecasting (SAFNWC) cloud and snow mask product to identify clear-sky conditions and exclude observations with large airmass (e.g., solar or viewing zenith angles >80°). In addition, observation geometry parameters, including solar zenith angle (SZA), view zenith angle (VZA), and scattering angle (SCA), were used to characterize satellite viewing conditions and better isolate aerosol signals.

We also collected the operational Level 2 aerosol product from GOES-16/ABI, which provides AOD at 550 nm with a spatial resolution of 2 km, generated using a physics-based, look-up-table-assisted multispectral retrieval method adapted from the MODIS/VIIRS Dark Target and ocean multi-channel heritage (GOES-R ABI AOD ATBD, 2018). In this study, hourly AOD retrievals at 550 nm over land that met the highest quality criteria (quality flag = 3) were used as a baseline reference for comparison with our deep-learning-based aerosol retrievals.

2.2. AERONET ground-based measurements

Aerosol Robotic Network (AERONET) provides a long-term (> 25 years), continuous, and publicly accessible database of aerosol optical properties across a standardized spectral range (440–1020 nm), with high temporal resolution (typically 15 min) (Giles et al., 2019). Data are available at three quality levels: Level 1.0 (unscreened), Level 1.5 (cloud-screened and quality-controlled), and Level 2.0 (fully cloud-screened, post-calibrated, and quality-assured). AERONET provides highly accurate AOD, with an uncertainty of ~0.01 for Level 2.0 product in the visible bands (Giles et al., 2019; Zhang et al., 2025). It also provides fine-mode and coarse-mode AOD (fAOD and cAOD) at 500 nm using the spectral deconvolution algorithm (SDA), with accuracy primarily determined by uncertainties in direct-sun measurements. Additionally, inversion products include key aerosol properties, such as SSA and AF, retrieved from multi-angle (sky-scanning) measurements and typically available at hourly temporal resolution.

In this study, quality-assured (Level 2.0) AOD at 440, 500, 675, 870, and 1020 nm, and fAOD and cAOD at 500 nm from AERONET Version 3.0 products, are used as references for model training and validation. These high-frequency measurements are averaged to hourly means to

match satellite observations. Due to the limited availability of Level 2.0 inversion products, hourly quality-controlled (Level 1.5) SSA and AF at 440, 675, 870, and 1020 nm are used instead, with additional screening criteria applied. The AOD > 0.3 criterion was applied because AERONET inversion products are generally less reliable under low-aerosol-loading conditions (Dubovik et al., 2000). The 440-nm SSA > 0.7 and sky-radiance fitting residual <10% criteria were used to exclude uncertain inversion results and improve retrieval quality (Jiang et al., 2024; Zhang et al., 2025). Notably, the residual uncertainty in the Level 1.5 products may introduce training noise and partly contribute to the lower retrieval accuracy for SSA and AF compared with AOD-related variables.

2.3. Auxiliary data

The Goddard Earth Observing System Composition Forecast (GEOS-CF) model is used to provide hourly aerosol composition data, including AOD at 550 nm for Organic Carbon (OC), Black Carbon (BC), Sea Salt (SAL), Mineral Dust (Dust), and Sulfate (SO_4). These species are simulated at 0.25 resolution within the GEOS-Chem chemical transport framework driven by NASA's GEOS forecast meteorology (Keller et al., 2021). To account for atmospheric gas absorption in multispectral satellite observations (Hsu et al., 2013; Wei et al., 2019), we additionally incorporated hourly total column ozone and precipitable water from the Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2) reanalysis ($0.5^\circ \times 0.625^\circ$ resolution) (Randles et al., 2017). Additional auxiliary datasets from the Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM; 90 m) for terrain characterization (Yang et al., 2011). All auxiliary variables are spatially harmonized with GOES-16 imagery, with higher-resolution datasets aggregated and lower-resolution datasets resampled using bilinear interpolation. All observations and ancillary data used for model training and evaluation were collected in 2021. In addition, observations from 2022, which were not used for model training, were included for independent validation.

3. Methods

3.1. Multi-parameter, multi-wavelength aerosol retrieval framework

This study develops a multi-task deep-learning framework for the hourly joint estimate of multiple aerosol properties from GOES-16/ABI observations and other auxiliary data. Unlike most previous studies, which typically retrieve a single aerosol parameter independently (Fan and Sun, 2023; Kang et al., 2022; Sun et al., 2025; Tang et al., 2025; Tao et al., 2023; Yan et al., 2024), our framework simultaneously retrieves five key aerosol properties, including multi-wavelength aerosol optical depth (AOD at 440, 500, 675, 870, and 1020 nm), fine-mode AOD (fAOD at 500 nm), coarse-mode AOD (cAOD at 500 nm), single scattering albedo (SSA at 440, 675, 870, and 1020 nm), and the asymmetry factor (AF at 440, 675, 870, and 1020 nm), using AERONET measurements as reference data:

$$AERONET\begin{bmatrix} AOD_{440}, AOD_{500}, AOD_{675}, AOD_{870}, AOD_{1020} \\ fAOD_{500}, cAOD_{500} \\ SSA_{440}, SSA_{675}, SSA_{870}, SSA_{1020} \\ AF_{440}, AF_{675}, AF_{870}, AF_{1020} \end{bmatrix} = AeroCatTrans(Inputs) \quad (1)$$

Where "Inputs" refers to the set of predictor variables used by the retrieval model, including satellite-observed spectral reflectance, observation geometry, meteorological variables, and surface-related information.

At the core of this framework is a deeply integrated model, AeroCatTrans, which combines a classical tree-based CatBoost machine-learning model with a Transformer-based deep-learning architecture to enable consistent and accurate multi-parameter retrieval (Fig. 1). In the first stage, CatBoost (Hancock and Khoshgoftaar, 2020) serves as a feature extractor and initial prediction module, ingesting dynamic inputs from multi-source observations. As a high-performance gradient boosting decision tree (GBDT) framework, it effectively handles heterogeneous inputs and multi-output regression tasks, providing robust first-order estimates of multiple aerosol parameters. Compared with other tree-based ensemble methods such as Random Forest and XGBoost, CatBoost employs gradient boosting to better capture complex nonlinear relationships among spectral, angular, and auxiliary variables. Its ordered boosting strategy helps reduce prediction bias and overfitting, thereby improving generalization performance.

In the second stage, a Transformer model (Vaswani et al., 2017) is employed to further learn latent representations and capture complex interdependencies among aerosol properties. Using a self-attention mechanism, the Transformer is particularly effective at modeling long-range dependencies and structured relationships in high-dimensional feature spaces, enabling refinement of initial estimates, reduction of residual errors, and enforcement of cross-parameter consistency. The final results are obtained by combining the outputs of both models with different weights (CatBoost at 0.3 and Transformer at 0.7) to improve accuracy and robustness.

3.1.1. Multi-band AOD retrieval

For aerosol loading retrieval, multi-wavelength AOD at 400 nm, 500 nm, 675 nm, 870 nm, and 1020 nm from AERONET are used as multi-task targets ($AOD^{AERONET}_{multi-band}$) in the AeroCatTrans model. In the first stage, CatBoost ingests hourly GOES-16/ABI TOA reflectance from five shortwave bands (0.45–2.2 μm), which are highly sensitive to aerosol extinction. To account for observation geometry (*Angles*), angular information including *SZA* and *VZA* is employed, while *SCA* is included to characterize the photon path length and aerosol scattering phase functions (Wei et al., 2024). In addition, GEOS-CF aerosol components ($GEOS_{AC}$: OC, BC, SAL, Dust, and SO_4) are incorporated as a prior constraint (Luffarelli and Govaerts, 2019; Wang et al., 2025a):

$$AOD^{AERONET}_{400,500,675,870,1020} \sim CatBoost(TOA_{0.45-2.2}, Angles, GEOS_{AC}) \quad (2)$$

In the second stage, auxiliary features, including total column ozone (*TCOZ*), total column precipitable water vapor (*TCPW*), digital elevation (*DEM*), spatial coordinates (*S*), temporal indicators (*T*) (Cheng et al., 2025), statistical summaries of CatBoost outputs (mean, maximum, and minimum; $\overline{AOD}^{CatBoost}_{statistical}$), and together with satellite observations ($TOA_{0.45-2.2}$ and *Angles*), are subsequently fed into the Transformer to refine the retrievals. Notably, the inputs to the Stage-2 Transformer are not the same as those used in Stage 1, nor are the original retrieval inputs directly passed into Stage 2. Instead, Stage 2 uses statistical summaries derived from the Stage-1 retrieval outputs, including the mean, maximum, and minimum values of the retrieved aerosol properties across spectral bands. These summarized retrieval features are then combined with additional satellite observations, meteorological and aerosol-related variables, and used as inputs to the Stage-2 Transformer:

$$AOD^{AERONET}_{multi-band} \sim Transformer\left(TOA_{0.45-2.2}, Angles, TCOZ, TCPW, DEM, S, T, \overline{AOD}^{CatBoost}_{statistical}\right) \quad (3)$$

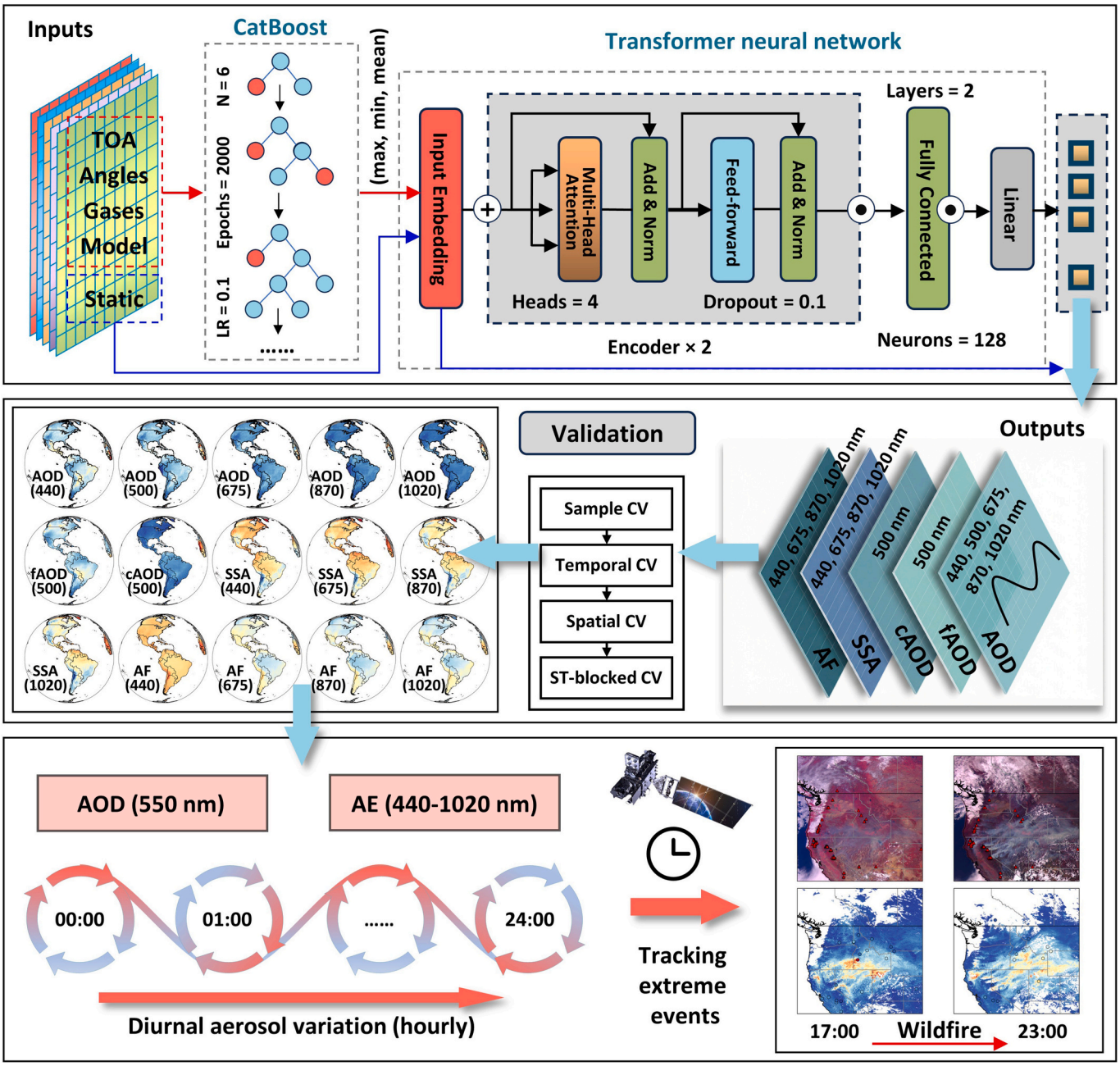

Fig. 1. Flowchart of the AeroCatTrans model for multi-parameter, multi-wavelength aerosol optical property retrieval from geostationary satellite observations (demonstrated using GOES-16/ABI).

The statistical AOD inputs provide important contextual information on the local aerosol background and spatial variability around each target pixel. These statistical descriptors complement the pixel-level features by enabling the Transformer to model local contextual relationships and spatial dependencies, thereby improving retrieval robustness.

3.1.2. *Fine- and coarse-mode aerosol retrieval*

For fine- and coarse-mode aerosol retrieval, AERONET-derived fAOD and cAOD at 500 nm are used as multi-task training targets ($fAOD_{500}^{AERONET}$ and $cAOD_{500}^{AERONET}$) in the AeroCatTrans model. In the first stage, CatBoost ingests hourly multi-band TOA reflectances (0.45–2.2 μm), angle information (*SZA*, *VZA*, and *SCA*), and GEOS-CF aerosol composition simulations ($GEOS_{AC}$):

$$\begin{bmatrix} fAOD_{500}^{AERONET} \\ cAOD_{500}^{AERONET} \end{bmatrix} \sim CatBoost(TOA_{0.45-2.2}, Angles, GEOS_{AC}) \quad (4)$$

In the second stage, auxiliary features, including total-column absorbing gases (*TCOZ* and *TCPW*), digital elevation (*DEM*), and spatiotemporal indicators (S, T), are incorporated. These variables, together with statistical summaries of CatBoost outputs, satellite multispectral TOA reflectance ($TOA_{0.45-2.2}$), and angle information, are fed into the Transformer to refine the fAOD and cAOD retrievals:

$$\begin{bmatrix} fAOD_{500}^{AERONET} \\ cAOD_{500}^{AERONET} \end{bmatrix} \sim Transformer\left(\begin{bmatrix} TOA_{0.45-2.2}, Angles, TCOZ, TCPW, DEM, S, T, \overline{fAOD}_{\text{statistical}}^{CatBoost} \\ TOA_{0.45-2.2}, Angles, TCOZ, TCPW, DEM, S, T, \overline{cAOD}_{\text{statistical}}^{CatBoost} \end{bmatrix} \right) \quad (5)$$

3.1.3. SSA and AF estimation

For multi-wavelength SSA and AF retrievals, AERONET measurements at 440, 675, 870, and 1020 nm are used as multi-task labels in the AeroCatTrans model. In the first stage, dynamic variables, including hourly TOA reflectances (0.45–2.2 μm), observation geometry, scattering angle (SCA), and aerosol components, are used as inputs to the CatBoost model, given their variability and their physical influence on aerosol scattering and absorption processes:

$$\begin{bmatrix} SSA_{400,675,870,1020}^{AERONET} \\ AF_{400,675,870,1020}^{AERONET} \end{bmatrix} \sim CatBoost(TOA_{0.45-2.2}, Angles, GEOS_{\text{AC}}) \quad (6)$$

In the second stage, in addition to the satellite observations, auxiliary features including absorbing-gases columns (*TCOZ* and *TCPW*), elevation (*DEM*), spatiotemporal variables (S, T), and statistical summaries of CatBoost outputs ($\overline{SSA/AF}_{\text{statistical}}^{CatBoost}$), are input to the Transformer to refine SSA and AF retrievals:

$$\begin{bmatrix} SSA_{400,675,870,1020}^{AERONET} \\ AF_{400,675,870,1020}^{AERONET} \end{bmatrix} \sim Transformer\left(\begin{bmatrix} TOA_{0.45-2.2}, Angles, TCOZ, TCPW, DEM, S, T, \overline{SSA}_{\text{statistical}}^{CatBoost} \\ TOA_{0.45-2.2}, Angles, TCOZ, TCPW, DEM, S, T, \overline{AF}_{\text{statistical}}^{CatBoost} \end{bmatrix} \right) \quad (7)$$

3.1.4. Model training and implementation

During model training, the CatBoost component is optimized using a multi-task root mean square error (RMSE) loss, with a tree depth of 6, a learning rate of 0.1, a maximum of 2000 iterations, and early stopping after 20 rounds. The second-stage Transformer encoder (Vaswani et al., 2017) is employed to capture complex temporal dependencies from the features generated by CatBoost. Each layer consists of multi-head self-attention and a position-wise feed-forward network, combined with residual connections and layer normalization.

Input features are projected into the model dimension through a linear embedding layer before being processed by the attention mechanism. Key hyperparameters, including the model dimension (128), number of attention heads (4), number of encoder layers (2), feed-forward dimension (512), dropout rate (0.1), and two fully connected layers with 128 hidden units each, are optimized empirically to balance predictive accuracy and computational efficiency. ReLU is applied as the activation function between hidden layers to enhance nonlinear representation capacity and promote sparse feature learning. The Transformer model is trained using mean squared error (MSE) loss to ensure stable optimization on a GPU (CUDA) (Fig. 1).

The loss weights for different retrieval targets (including different aerosol parameters and wavelengths) were determined based on the variance of each target in the training data. This helps balance the contributions of different outputs during training and reduces potential bias in the multi-task learning process. The total training time is 338.49 s, with a peak GPU memory usage of 79.70 MB. The inference time is 0.05 s for 16,061 samples, with a peak GPU memory usage of 94.17 MB. The execution time was measured using wall-clock time with CUDA synchronization, and GPU memory usage was tracked using PyTorch CUDA peak memory statistics.

Notably, the proposed framework is designed for high-quality aerosol retrieval rather than near-real-time operational applications. Therefore, it incorporates auxiliary reanalysis and model information that may not be available on a real-time basis. All input variables are used at the pixel level and are extracted from the nearest-neighbor satellite pixel corresponding to each AERONET site. No spatial aggregation (e.g., averaging over an 11 × 11-pixel window) was applied. This approach preserves the native spatial information of the satellite observations and ensures consistency between the satellite observations and the corresponding ground-based measurements.

We further conducted comparisons with the individual CatBoost and Transformer models. The proposed hybrid model consistently outperforms both baseline models across different aerosol optical properties (Table S2), demonstrating the advantage of integrating CatBoost and Transformer for aerosol retrieval.

3.2. Calculation of additional aerosol parameters

AOD at 550 nm (a commonly used reference wavelength) is derived from our multi-wavelength AOD retrievals (440, 500, 675, 870, and 1020 nm) using a robust second-order polynomial fitting approach (Wei et al., 2019):

$$\log(AOD_{550}) = a_0 \log(550) + a_1 \log(550)^2 + b \quad (8)$$

where a_0, a_1, and b are regression coefficients.

The Ångström Exponent (AE) for 440–1020 nm is estimated by least-squares fitting of a log-log linear relationship to multi-wavelength AOD_λ retrievals (where λ represents wavelengths of 440, 500, 675, 870, 1020 nm) from the AeroCatTrans model:

$$\log(AOD_\lambda) = -AE \log(\lambda) + b \quad (9)$$

The biomass burning index (BBI) is selected to enhance and identify aerosol dominated by wildfire emissions, and is computed from AE of 440–1020 nm ($AE_{440-1020}$) when the AOD at 440 nm exceeds 0.5:

$$\text{BBI_AE} = \frac{1}{1 + e^{-k(AE_{440-1020} - C_1)}} \quad (10)$$

where *k is* slope parameters that control the steepness of the curve (set to 10), and C_1 is threshold parameters with values 1.7 (Bibi et al., 2016; Kedia et al., 2014; Mielonen et al., 2009).

An alternative BBI is calculated as the product of RAOD (the ratio of AOD at 1020 nm to AOD at 440 nm) and AOD_{440}:

$$\text{BBI_RAOD} = AOD_{440} \times \text{RAOD} \quad (11)$$

where, if RAOD <0.25, it is set to 1; otherwise, it is set to 0.01 (Chen et al., 2016).

3.3. Validation strategies

We validated the hourly multi-band aerosol optical properties retrieved by the AeroCatTrans model using the ten-fold cross-validation (CV) approach (Fig. 1). First, we performed random ten-fold CV by partitioning all samples into ten folds and iteratively training on nine folds (90%) while validating on the remaining fold (10%) to generate out-of-fold predictions as a baseline. This procedure was repeated 10 times so that each fold served once as the validation set. To assess temporal generalization when ground observations are unavailable, we also conducted two temporal ten-fold CV approaches by splitting samples into ten folds based on measurement hour and day of year (DOY), respectively, and repeating the same training-validation cycle. To reduce uncertainty due to the uneven global distribution and clustering of ground sites, we performed a spatial ten-fold CV in which folds were defined by station locations. We also implemented a stricter spatiotemporal-blocked cross-validation framework, in which training samples were excluded if they were located within 50 km of the validation sites or fell within the same DOY-based temporal block as the validation samples. The total number of samples ranges from 58,915 to 160,604 across different aerosol properties, covering 16 daytime hours and 162–183 AERONET sites in 2021. In addition, observations from 2022, which were not used for model training, were included for independent validation.

The cross-validation correlation coefficient (R), root mean square error (RMSE), and mean absolute error (MAE) were used to quantitatively evaluate model accuracy and retrieval uncertainty for each

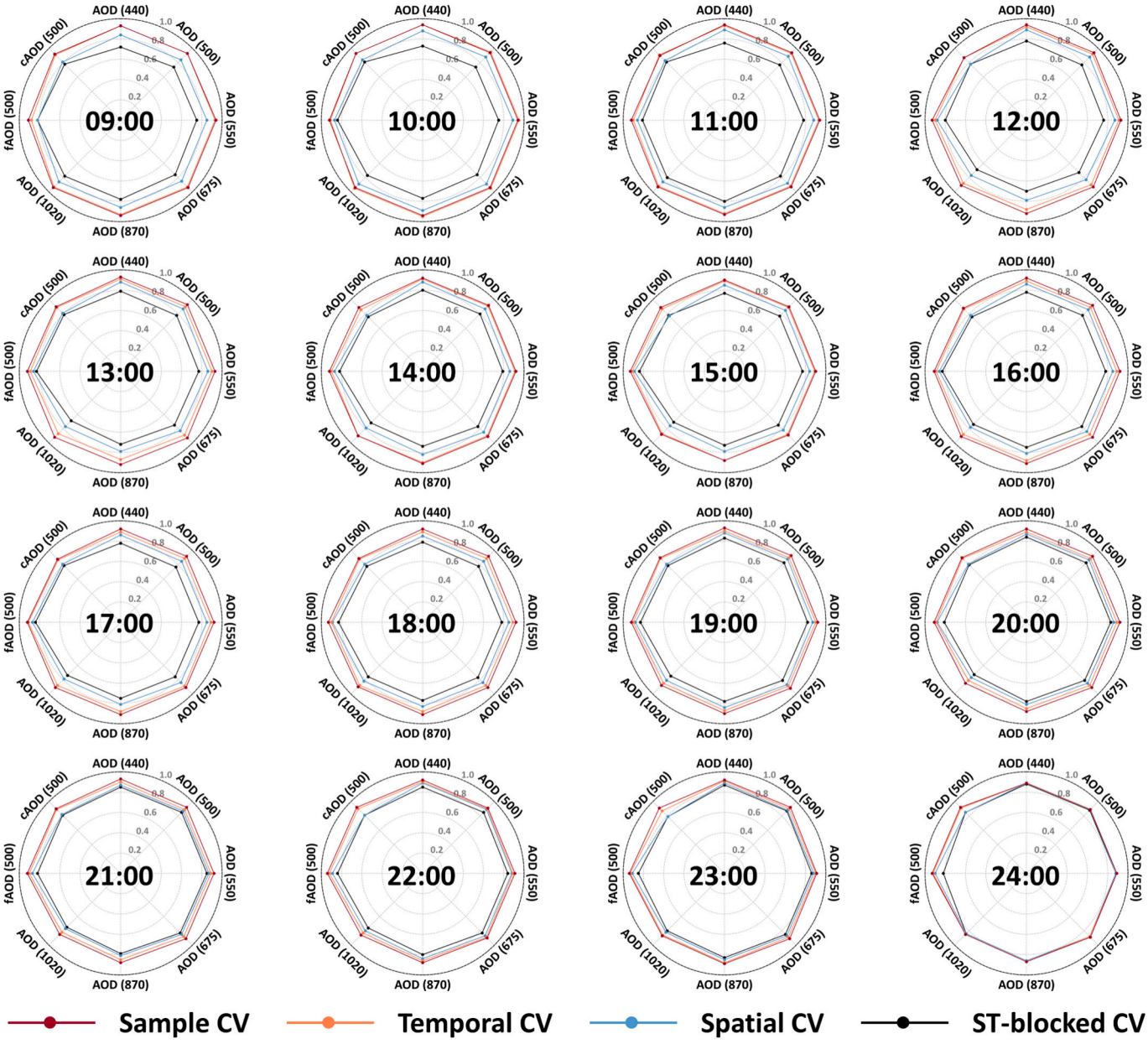

Fig. 2. Diurnal variation in retrieval accuracy (R) of hourly aerosol optical depth (AOD) across multiple wavelengths (440, 500, 550, 675, 870, and 1020 nm), as well as fine-mode AOD (fAOD; 500 nm) and coarse-mode AOD (cAOD; 500 nm), from 09:00 to 24:00 UTC, assessed using ten-fold sample-based (red lines), temporal (DOY-based; orange lines), spatial (site-based; blue lines), and spatiotemporal-blocked (black lines) cross-validation approaches. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

aerosol optical property. The reliability of aerosol loading retrievals was additionally assessed against the MODIS Deep Blue expected error (EE) envelope, defined as $\pm (0.05 + 20\% \times AOD_{\tau})$ (Hsu et al., 2013), as well as the stricter Global Climate Observing System (GCOS) requirement, defined as $\pm$ max(0.03, 10% × AOD_{τ}). Multi-band SSA were also evaluated using the GCOS criterion, defined as ±0.02 (GCOS, 2010). As no GCOS criterion is available for AF, the criterion used in this study was defined as ±0.02 for evaluation purposes.

4. Results and discussion

4.1. Validation of hourly aerosol property retrievals

4.1.1. Aerosol optical depth and size modes

Fig. 2 shows the diurnal variation in the accuracy of hourly AOD retrievals across multiple wavelengths (440–1020 nm) and aerosol size modes (fine- and coarse-mode AOD), evaluated using different cross-validation approaches. AeroCatTrans demonstrates stable performance in retrieving hourly AOD from GOES-16/ABI imagery, with sample-based CV-R values consistently ranging from 0.89 to 0.94 from 09:00 to 24:00 UTC. Across wavelengths, shorter wavelengths (440–675 nm) exhibit the highest and most stable correlations throughout the day across different CV approaches. In contrast, longer wavelengths (870 nm and 1020 nm) show systematically lower correlations, with the reduction most pronounced for spatial CV. This pattern is likely associated with reduced signal sensitivity and a narrower dynamic range of AOD at longer wavelengths.

In terms of diurnal variation, retrieval accuracy remains relatively stable from morning to afternoon local time (approximately 12:00–20:00 UTC), when all wavelengths and CV schemes reach peak or near-peak performance. Slight decreases are observed during early morning (09:00–11:00 UTC) and later in the day (after ~20:00 UTC), indicating increased uncertainty under less favorable illumination or atmospheric conditions. For aerosol modes, both fAOD and cAOD (500 nm) follow similar diurnal patterns to total AOD. However, fAOD generally achieves higher correlations than cAOD across all hours and validation schemes, while cAOD exhibits slightly larger variability, especially under spatial CV, indicating lower spatial generalization.

Across all wavelengths (440–1020 nm), the retrieved AOD exhibits strong agreement with ground-based observations, with high correlation coefficients R of 0.90–0.92 and regression slopes of 0.75–0.82 across bands (Fig. S1). Retrieval uncertainties decrease at longer visible and near-infrared wavelengths, reflected in lower RMSE and MAE values, primarily due to reduced aerosol optical depth and weaker signal sensitivity. The fraction of retrievals within the EE envelope exceeds 89% for all wavelengths and rises with wavelength, reaching 97% at 870–1020 nm. Similarly, the percentage of retrievals meeting GCOS requirements increases from 57% at 440 nm to 84% at 1020 nm. For the widely used interpolated 550 nm band, the model demonstrates high performance ($R = 0.92$, RMSE = 0.078, and MAE = 0.037), with 92.58% of retrievals falling within the EE envelope and 66.28% meeting the GCOS requirements. In addition, over 65% of stations exhibit both high correlations ($R > 0.7$) and low errors (RMSE <0.2) across all bands (Fig. S2), confirming the reliability of AeroCatTrans for multi-band AOD

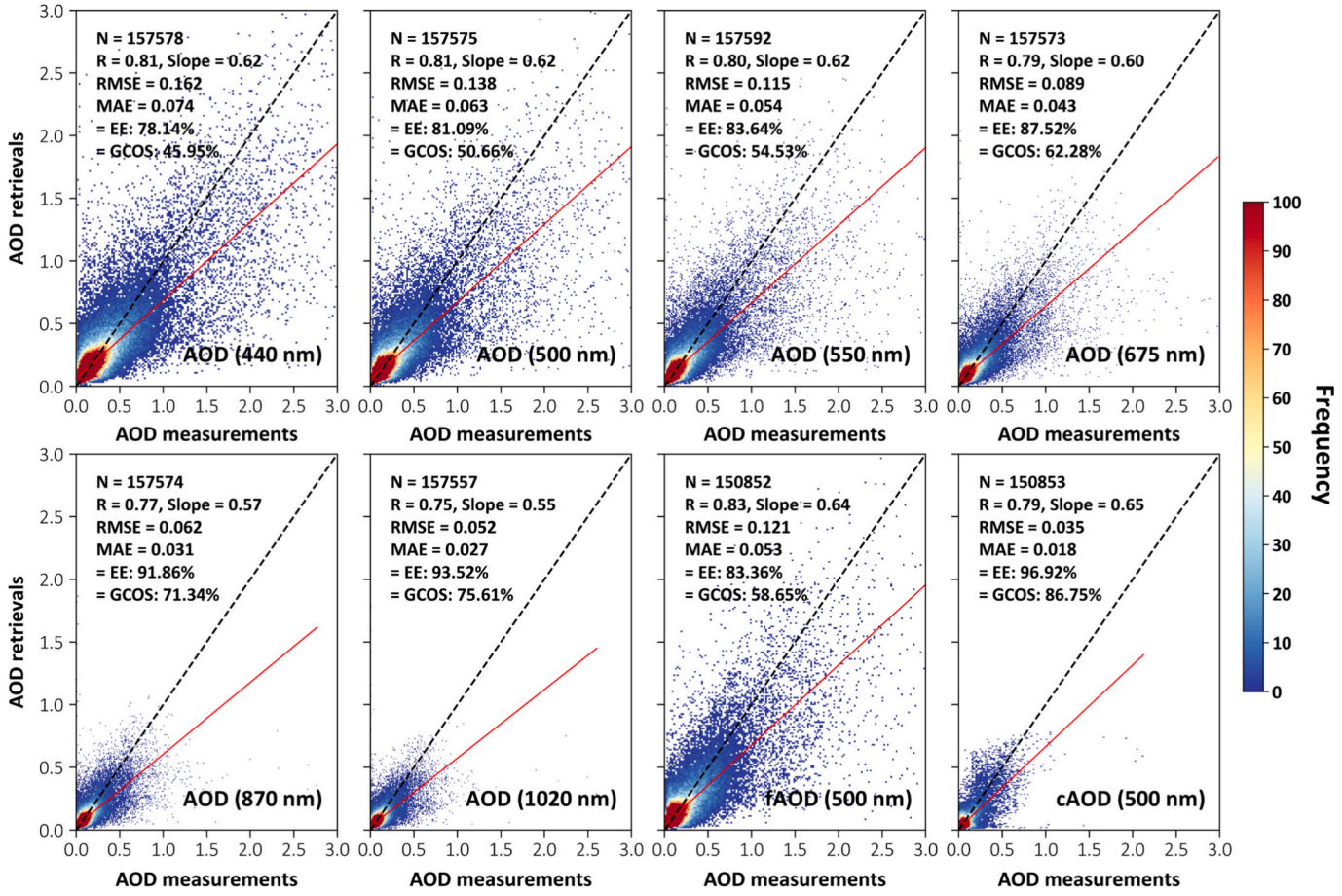

Fig. 3. Density scatter plots comparing hourly aerosol retrievals with AERONET measurements under the spatiotemporal-blocked cross-validation framework for AOD at multiple wavelengths (440, 500, 550, 675, 870, and 1020 nm), as well as fine-mode AOD (fAOD; 500 nm) and coarse-mode AOD (cAOD; 500 nm). The dashed black lines represent the 1:1 reference lines, and the red lines represent the fitted linear regression lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

retrieval.

For aerosol size modes, fAOD (500 nm) shows strong performance ($R = 0.93$, RMSE = 0.082, MAE = 0.036), comparable to total AOD (500 nm) but with slightly larger spread at higher values, reflecting increased uncertainty in separating aerosol components. Overall, 91.97% of retrievals fall within the EE envelope, and 69.69% meet GCOS requirements. In contrast, cAOD (500 nm) exhibits lower correlation ($R = 0.89$) but substantially smaller absolute errors (RMSE = 0.026, MAE = 0.014), likely due to its lower magnitude and narrower dynamic range. Notably, cAOD achieves the highest fraction within EE (98%) and GCOS compliance (91%), indicating more stable but lower-variability retrievals. Spatially, over 65% of stations show reliable hourly fAOD retrievals ($R > 0.7$, RMSE <0.2) and over 70% show reliable cAOD retrievals (e.g., $R > 0.5$, RMSE <0.1), further supporting the overall high accuracy of the model.

When specific hours are withheld from training, the model remains robust for hourly AOD retrievals during 09:00–24:00 UTC, with only slight decreases in performance, showing CV-R values of 0.91–0.92 and RMSEs of 0.032–0.109 across bands (Fig. S3). With increasing time intervals, the model still achieves reliable multi-band AOD retrievals, with DOY-based CV-R values of 0.88–0.90 despite a slight decline in performance (Fig. S4). When entire sites are withheld, performance is lower than in sample-based and hour-based CV, yet the model still achieves high CV-R values of 0.82–0.88 and low RMSEs of 0.045–0.130 across bands, indicating good spatial generalization capability (Fig. S5). Similar patterns are observed for aerosol size modes: fAOD attains a temporal CV-R of 0.90–0.92 (RMSE = 0.085–0.092; Figs. S3-S4) and a site-based CV-R of 0.88 (RMSE = 0.102; Fig. S5), while cAOD shows strong temporal generalization (CV-R of 0.88–0.89, RMSE = 0.027; Figs. S3-S4) and moderate spatial generalization (CV-R of 0.80,

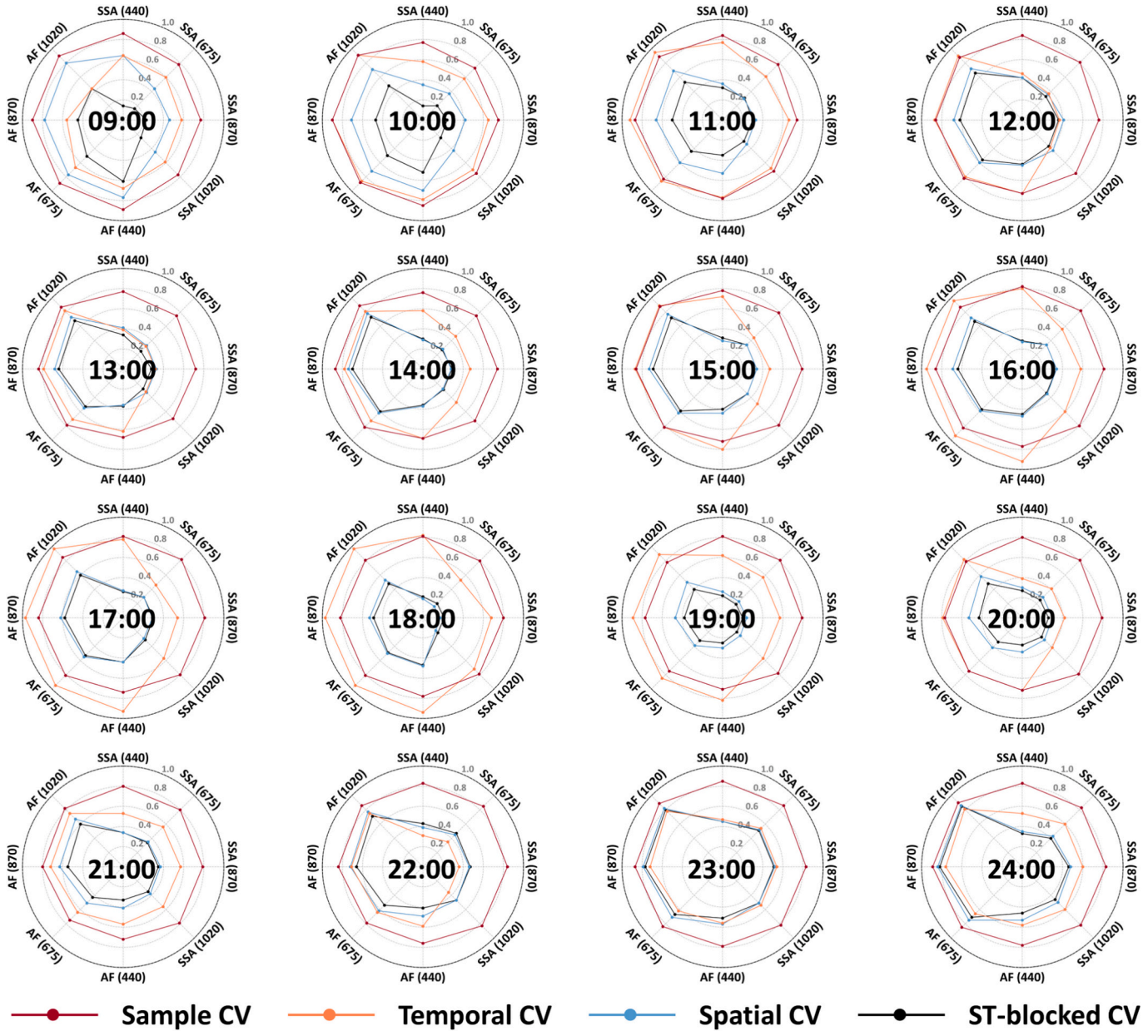

Fig. 4. Diurnal variation in retrieval accuracy (R) of hourly single-scattering albedo (SSA) and asymmetry factor (AF) across multiple wavelengths (440, 675, 870, and 1020 nm) from 09:00 to 24:00 UTC, assessed using ten-fold sample-based (red lines), temporal (DOY-based; orange lines), spatial (site-based; blue lines), and spatiotemporal-blocked (black lines) cross-validation approaches. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

RMSE = 0.035; Fig. S5).

The stricter spatiotemporal-blocked CV results show that the model maintains strong performance in retrieving multi-band AOD (440–1020 nm), with correlations of 0.75–0.81 and RMSE values of 0.052–0.162 (Fig. 3). High accuracy is also achieved for fAOD ($R = 0.83$, RMSE = 0.121) and cAOD ($R = 0.79$, RMSE = 0.035), respectively. Independent validation in 2022 further indicates the model's high robustness and extrapolation ability, with strong correlations to ground-measured multi-band AOD (0.81–0.84), fAOD (0.77), and cAOD (0.86) (Fig. S6).

4.1.2. *Single-scattering albedo and asymmetry factor*

Fig. 4 illustrates the diurnal variation in retrieval accuracy of hourly SSA and AF across multiple wavelengths (440–1020 nm). The inference of aerosol microphysical properties is inherently more challenging than that of AOD, primarily due to the limited sensitivity of TOA reflectance to these parameters, as well as their strong dependence on aerosol and surface conditions (Bao et al., 2024; Chen et al., 2020). Nevertheless, AeroCatTrans achieves consistent performance in SSA retrieval throughout the daytime, with moderate correlations under sample-based CV ($R > 0.5$) across all wavelengths between 10:00 and 24:00 UTC. Retrieval accuracy is broadly comparable among wavelengths, with the 440 nm band showing slightly better performance. In contrast, DOY-based CV exhibits lower accuracy, with reduced R values (0.34–0.69) compared with the sample-based results (0.62–0.80). Spatial CV yields systematically lower correlations (0.18–0.50) across all hours. Temporally, the model shows stable performance during most daytime hours, while retrieval uncertainties are generally higher during the early morning (09:00–11:00 UTC) and late afternoon (after ~20:00 UTC), particularly in terms of spatial generalization.

The AeroCatTrans model also demonstrates stable performance in retrieving hourly AF, with sample-based CV-R values ranging from 0.75 to 0.90 between 10:00 and 24:00 UTC, generally outperforming SSA retrievals. Across all wavelengths, longer wavelengths (870 nm and 1020 nm) exhibit higher and more stable correlations ($R = 0.75$–0.94) across all three CV approaches, whereas shorter wavelengths (440 and 675 nm) show lower accuracy, with the most pronounced degradation in spatial generalization. Diurnally, retrieval accuracy remains relatively stable from late morning to early afternoon (10:00–20:00 UTC), when all wavelengths and validation schemes reach peak or near-peak performance. Reduced accuracy is observed in the early morning (09:00 UTC) and late afternoon (after 21:00 UTC), particularly at shorter wavelengths, where spatial generalization is lower.

Overall, hourly SSA retrievals show moderate agreement with ground-based observations, with the best performance at 440 nm, where the sample-based CV-R reaches 0.63, decreasing to around 0.50 at other wavelengths (Fig. S7). Retrieval uncertainty increases toward longer wavelengths, as indicated by higher RMSE and MAE values. The proportion of retrievals meeting GCOS accuracy requirements is approximately 46%–68% across all wavelengths. Only 11%–35% of sites exhibit both moderate correlations ($R > 0.5$) and low RMSE (< 0.05) across bands (Fig. S8), underscoring the persistent challenge of accurately retrieving SSA. In addition, the model demonstrates consistently good performance in retrieving SSA when specific hours or days are withheld from training, maintaining stable results across wavelengths (temporal CV-$R = 0.44$–0.70; RMSE = 0.023–0.039) (Figs. S9-S10). However,

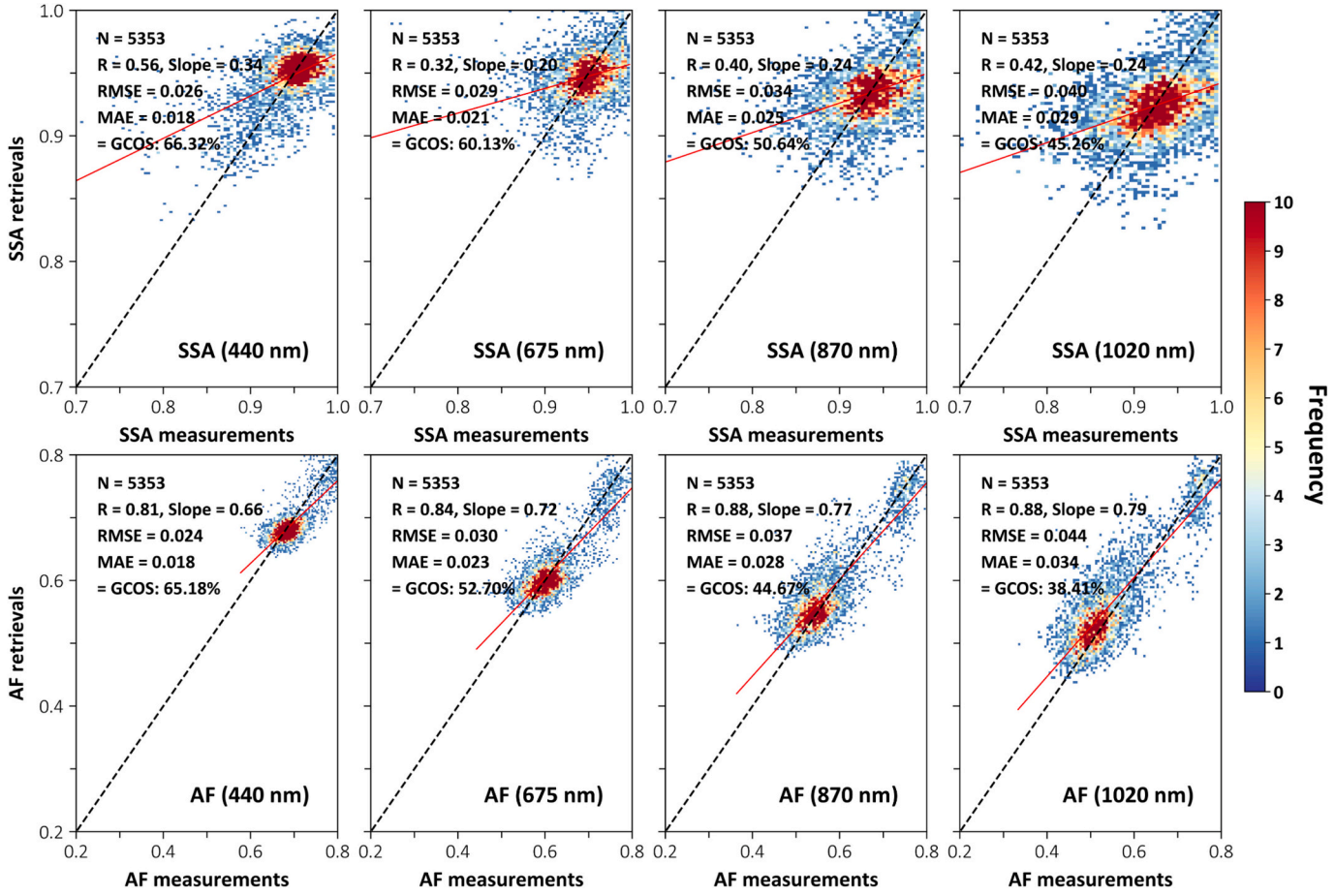

Fig. 5. Density scatter plots comparing hourly aerosol retrievals with AERONET measurements under the spatiotemporal-blocked cross-validation framework for SSA and AF across multiple wavelengths (440, 675, 870, and 1020 nm). The dashed black lines represent the 1:1 reference lines, and the red lines represent the fitted linear regression lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

performance declines substantially at unseen sites as reflected in lower site-based CV-R values (0.25–0.51) and higher RMSEs (0.029–0.045) (Fig. S11). This is likely due to the greater challenge of generalizing SSA retrievals from GOES-16/ABI multispectral observations.

Hourly AF retrievals exhibit substantially higher accuracy than SSA, with consistently strong sample-based CV-R values of 0.85–0.91 and regression slopes of 0.69–0.83 across wavelengths (Fig. S7). Retrieval uncertainty generally decreases with increasing wavelength, with the average RMSE declining from 0.036 to 0.021, while the proportion of retrievals meeting GCOS accuracy requirements decreases correspondingly from 72.33% to 46.14%. Over 65% of sites yield good AF retrievals, achieving both high CV-R (> 0.5) and low RMSEs (< 0.05) (Fig. S8). For temporal generalization, the model maintains stable retrieval performance, with strong hour (DOY)-based CV-R values of 0.85–0.91 (0.86–0.92) and low RMSEs of 0.021–0.036 (0.020–0.034) across wavelengths (Figs. S9-S10), indicating reliable predictions even in the absence of coincident ground observations. For spatial generalization, the model remains robust, with slightly reduced site-based CV-R values of 0.84–0.90 and RMSEs of 0.022–0.037 (Fig. S11), demonstrating reliable multi-wavelength AF retrieval capability in regions lacking ground observations.

The performance of multi-band SSA and AF decreases under the stricter spatiotemporal validation framework (SSA: $R = 0.32$–0.56, RMSE $= 0.026$–0.040; AF: $R = 0.81$–0.88, RMSE $= 0.024$–0.044) (Fig. 5). Fig. S12 shows the performance of the SSA retrievals from spatiotemporal-blocked CV as a function of retrieved AOD. As expected, SSA retrievals show relatively limited skill under low-AOD conditions (AOD < 0.3), with relatively large RMSEs and weak correlations with AERONET measurements. As retrieved AOD increases, the RMSE generally decreases across all four wavelengths, while the correlations tend to improve, particularly under moderate-to-high aerosol loading conditions (AOD > 0.5). These results confirm the expected AOD dependence of SSA retrieval performance, reflecting the stronger aerosol signal available to constrain SSA at higher aerosol loadings. They also indicate that our SSA product is more reliable under moderate-to-high AOD conditions, whereas greater caution is warranted under clean conditions. Nevertheless, the model retains meaningful predictive skill under this rigorous evaluation framework. This is further supported by the model's satisfactory predictive performance in 2022, when no data were used for model training, with R values of 0.25–0.51 for SSA and 0.84–0.90 for AF (Fig. S13).

4.2. Diurnal variations in aerosol optical properties

4.2.1. Aerosol optical depth (AOD)

Fig. 6 illustrates the diurnal evolution of the spatial distribution of aerosol optical depth (AOD) across the Americas at five wavelengths (440–1020 nm) from 10:00 to 24:00 UTC in 2021. AOD decreases systematically with increasing wavelength, with the highest values observed at 440 nm (0.17–0.27) and the lowest at 1020 nm (0.05–0.08). This spectral dependence is consistent with the dominance of fine-mode aerosols and their stronger scattering efficiency at shorter wavelengths. Elevated AOD is primarily concentrated over central South America (including the Amazon Basin), extending northward through Central America into southern Mexico, forming a prominent high-AOD belt across the tropical Americas. Additional enhancements are evident over

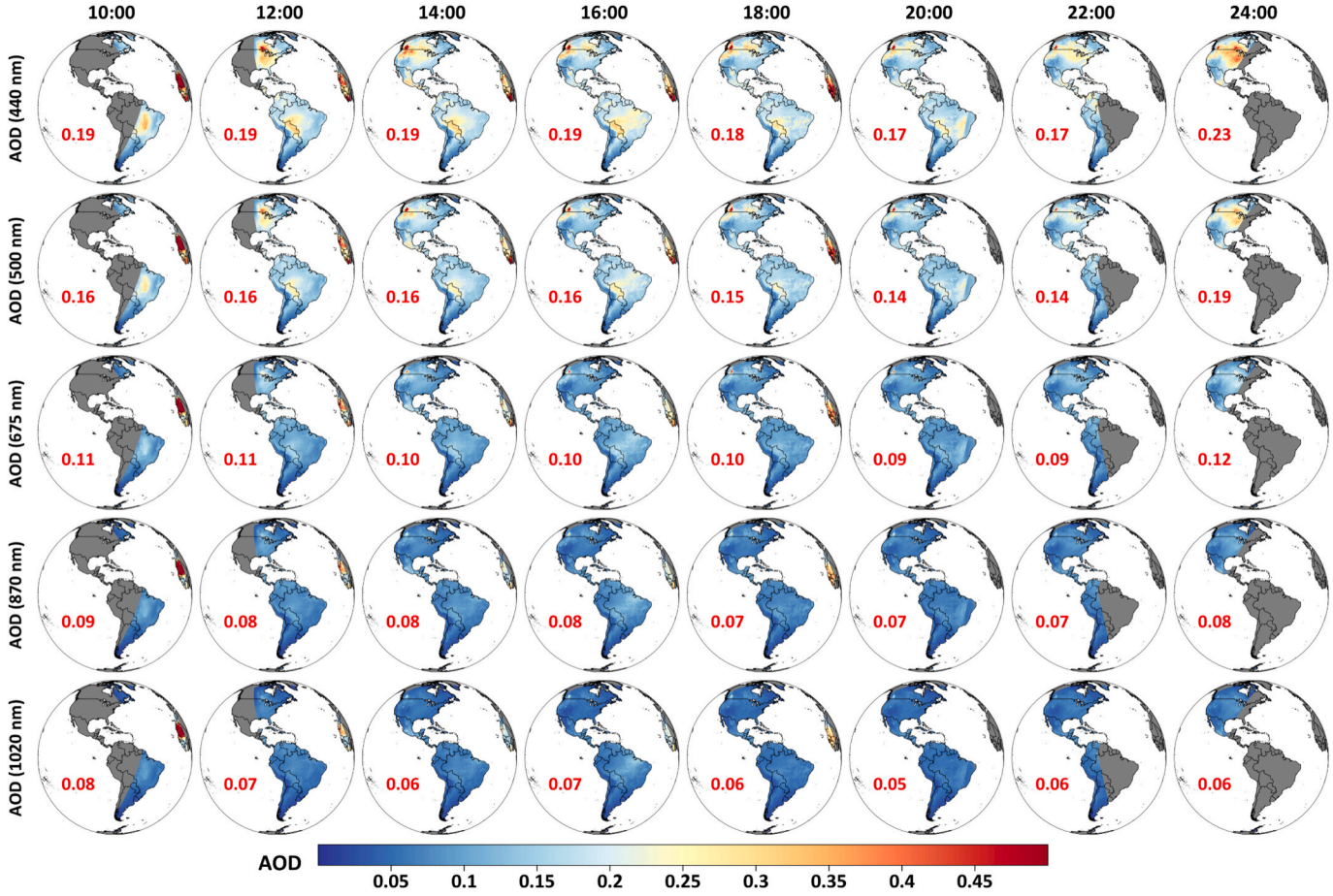

Fig. 6. Diurnal variations in the spatial distribution of aerosol optical depth (AOD) at 440, 500, 675, 870, and 1020 nm across the Americas from 10:00 to 24:00 UTC in 2021, retrieved using the AeroCatTrans model. Inset red numbers indicate the spatially averaged AOD at each hour. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

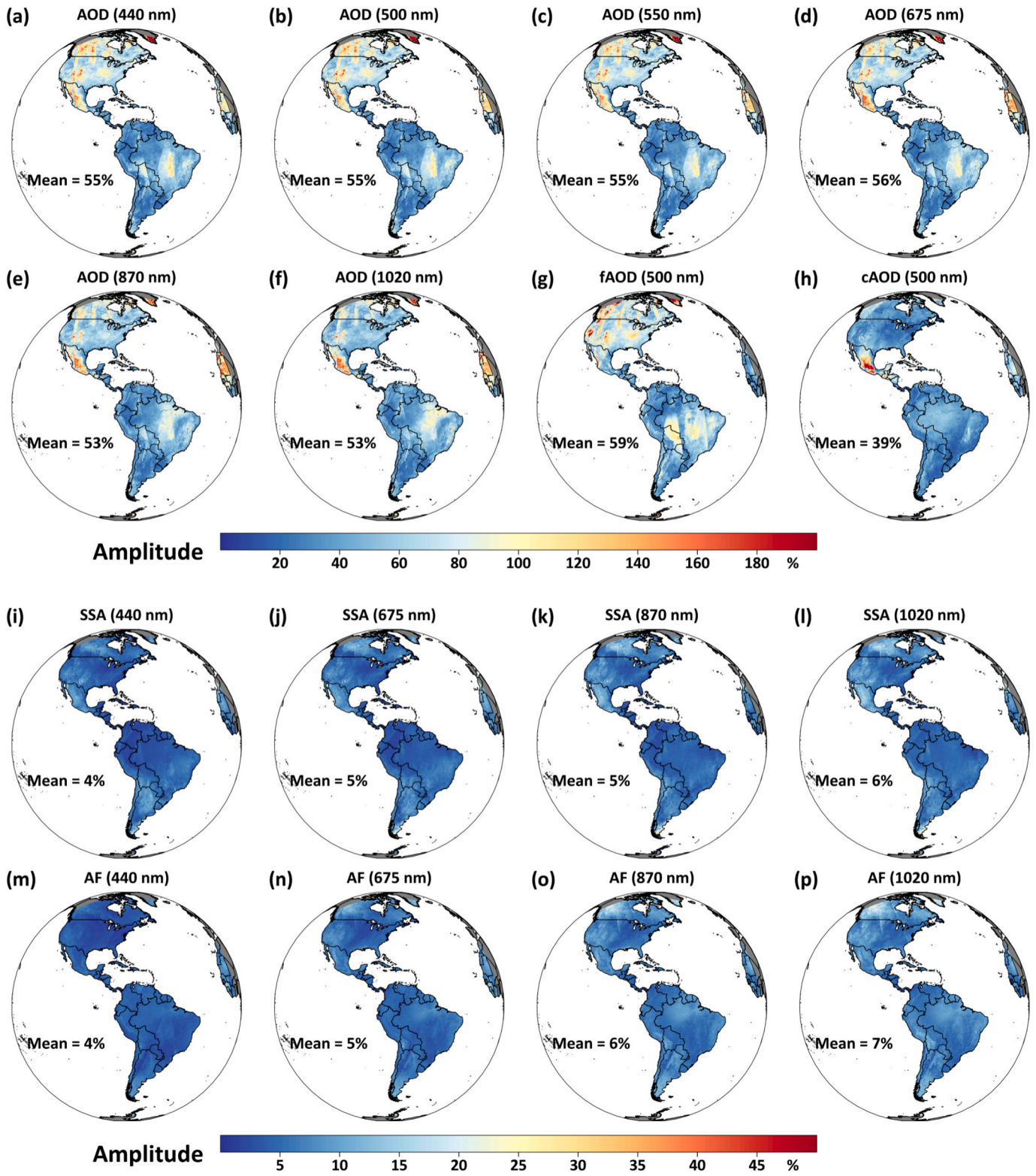

Fig. 7. Diurnal amplitude of aerosol parameters across the Americas in 2021: (a-f) multi-band aerosol optical depth (AOD), (g-h) fine-mode AOD (fAOD) and coarse-mode AOD (cAOD) at 500 nm, (i-l) single-scattering albedo (SSA), and (m-p) asymmetry factor (AF). Inset red numbers indicate mean values. The diurnal amplitude is defined as the difference between the maximum and minimum daytime values, normalized by the daytime mean. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the eastern United States and, intermittently, the western United States, reflecting contributions from biomass burning, biogenic emissions, anthropogenic sources, and regional transport. In contrast, relatively low AOD values are observed over high-latitude regions and cleaner marine environments.

In terms of diurnal variation, AOD exhibits a coherent temporal pattern across all wavelengths, with mean values generally decreasing from morning to afternoon (10:00–22:00 UTC). This pattern is

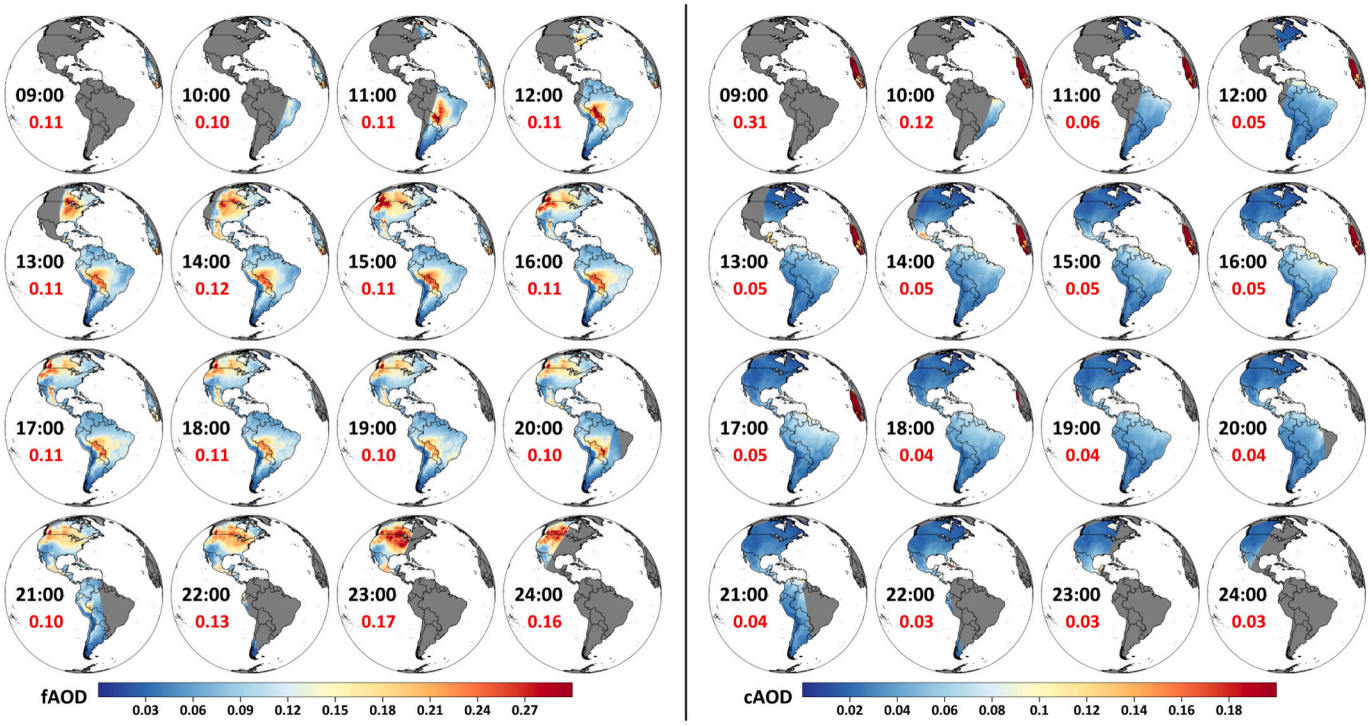

Fig. 8. Diurnal variations in the spatial distribution of fine-mode aerosol optical depth (550 nm) and coarse-mode aerosol optical depth (500 nm) from 09:00 to 24:00 UTC across the Americas in 2021, retrieved using the AeroCatTrans model. Inset black and red numbers indicate the hours and mean values, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

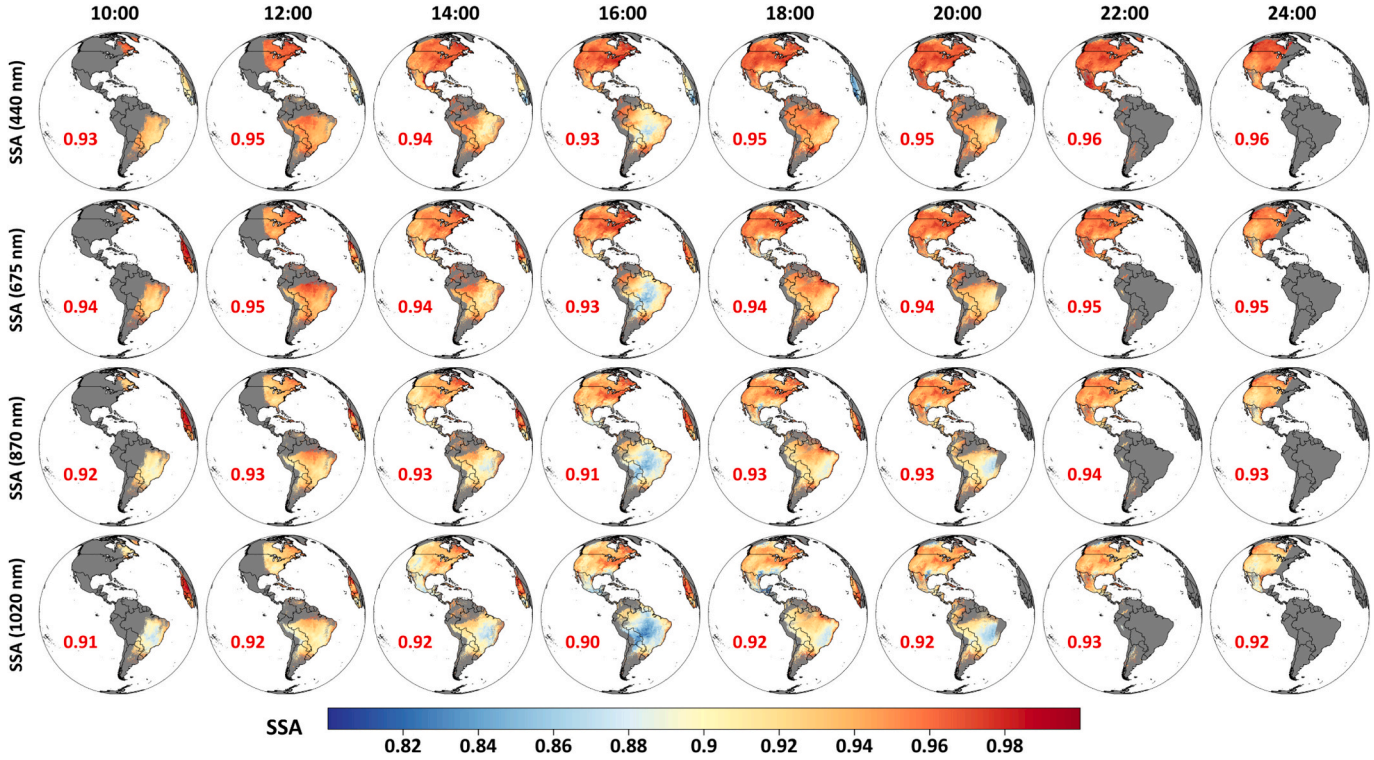

Fig. 9. Diurnal variations in spatial distribution of single-scattering albedo (SSA) at 440, 675, 870, and 1020 nm from 10:00 to 24:00 UTC across the Americas in 2021, retrieved using the AeroCatTrans model. Inset red numbers indicate the spatially averaged SSA at each hour. Only pixels with hourly AOD (550 nm) > 0.5 are included in the analysis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

particularly evident over regional domains, such as the United States, Gulf of Mexico, and southern South America. A similar but less pronounced trend is observed at longer wavelengths (e.g., 1020 nm). This daytime pattern likely arises from the interplay of boundary layer dynamics, aerosol emissions, and variations in solar illumination. The amplitude of diurnal variation is also more pronounced at shorter

wavelengths (440–675 nm), reaching 55%–56%, particularly over northwestern North America, indicating stronger temporal variability in fine-mode aerosols, likely associated with wildfire emissions (Fig. 7). In contrast, longer wavelengths (870–1020 nm) exhibit weaker variability, with amplitudes of 53%, and more spatially homogeneous patterns.

Consistent with the multi-band AOD results, the modal decomposition further reveals distinct diurnal behaviors of fine- and coarse-mode aerosols (Fig. 8). The fine-mode AOD (fAOD) exhibits a highly localized spatial distribution, with pronounced hotspots (>0.15) over central South America, particularly the southern Amazon and Pantanal region (Wei et al., 2023), as well as northwestern North America. In contrast, most other regions exhibit relatively low fAOD, highlighting the strong spatial heterogeneity and source-driven characteristics of fine-mode aerosols. fAOD also exhibits pronounced diurnal variability, especially over biomass-burning regions, with a domain-mean diurnal amplitude of 59% (Fig. 7g). Even stronger variability is observed over northwestern North America, where local amplitudes can exceed 200%.

In contrast, the coarse-mode AOD (cAOD) exhibits a more spatially diffuse and smoother distribution than fAOD, with the highest values over northern Africa (Sahara/Sahel) and the downwind tropical Atlantic dust outflow corridor (Fig. 8). Unlike the localized hotspots seen in fAOD, cAOD forms broad, continuous plumes associated with dust-dominated regions. In terms of temporal variability, cAOD shows relatively weak diurnal variation from 09:00 to 24:00 UTC, with a lower domain-mean amplitude of 39%, indicating limited sub-daily variability of coarse particles (Fig. 7h). Nevertheless, some regions, such as southern North America and northwestern Africa, exhibit locally enhanced variability, with amplitudes reaching 100%.

4.2.2. *Single-scattering albedo (SSA) and asymmetry factor (AF)*

Fig. 9 illustrates the diurnal variations in reliable SSA (550-nm AOD > 0.5) across the Americas at four wavelengths from 10:00 to 24:00 UTC in 2021. Spatially, high SSA values (> 0.92) are primarily observed over northern North America, particularly the eastern United States at shorter wavelengths (e.g., 440 nm), consistent with the prevalence of highly scattering aerosols, including secondary inorganic aerosols. Similarly elevated SSA values are found across much of tropical South America, suggesting the predominance of relatively scattering aerosol components. Relatively high SSA values are also evident over northwestern Africa, particularly at longer wavelengths, likely reflecting the increasing contribution of coarse-mode Saharan dust, which remains predominantly scattering despite its moderate absorptivity. In contrast, lower SSA values (indicating enhanced absorption) are observed in regions likely influenced by wildfire emissions and other absorbing aerosol components, notably over western North America and southern South America, where absorbing aerosols contribute to reduced SSA. Across the study domain, SSA generally decreases with increasing wavelength, with regional mean values declining from approximately 0.94 at 440 nm to 0.92 at 1020 nm, indicating relatively stronger aerosol absorption at longer wavelengths. Diurnally, SSA exhibits only minor temporal variability, with domain-averaged SSA varying by less than 0.01 throughout the day and a limited diurnal amplitude of less than 6% (Fig. 7i-l), underscoring the stability of aerosol absorption properties throughout the day.

Unlike SSA, AF retrievals (550-nm AOD > 0.3) exhibit more spatially diffuse and smoother distributions, with the highest values over northwestern Africa across all wavelengths, consistent with enhanced forward scattering by coarse-mode dust (Fig. 10). Elevated AF values are also observed along the Andes and over southern South America, particularly at shorter wavelengths (440 nm), likely reflecting the predominance of coarse-mode aerosols and relatively large particle sizes. In contrast, lower AF values are observed over central South America, as well as eastern Canada and eastern North America, indicating a greater contribution from fine-mode aerosols. Across the study domain, AF generally decreases with increasing wavelength, with regional mean values declining from approximately 0.69 at 440 nm to 0.58 at 1020 nm, indicating weaker forward scattering at longer wavelengths. The different spectral behaviors of AF mainly reflect differences in aerosol

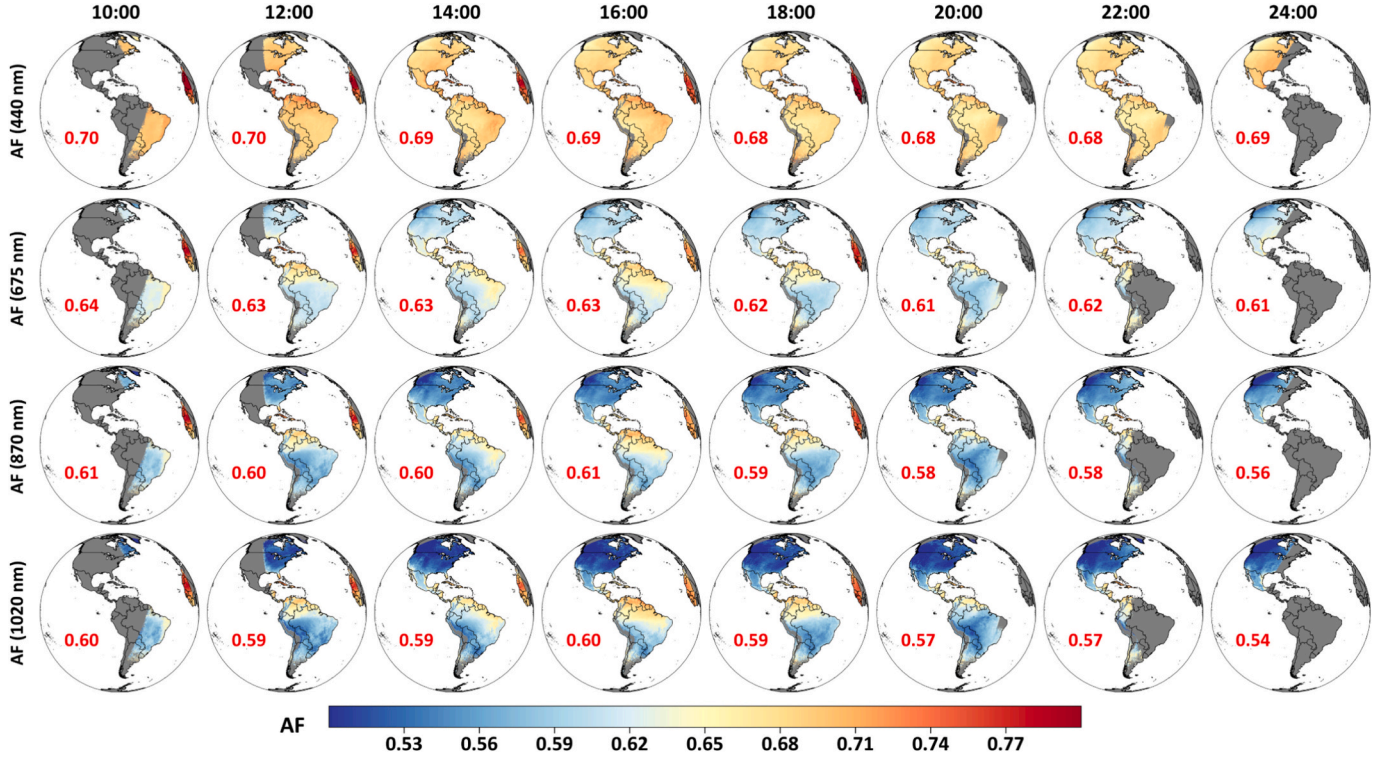

Fig. 10. Diurnal variations in the spatial distribution of asymmetry factor (AF) at 440, 675, 870, and 1020 nm from 10:00 to 24:00 UTC across the Americas in 2021, retrieved using the AeroCatTrans model. Inset red numbers indicate the spatially averaged AF at each hour. Only pixels with hourly AOD (550 nm) > 0.3 are included in the analysis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

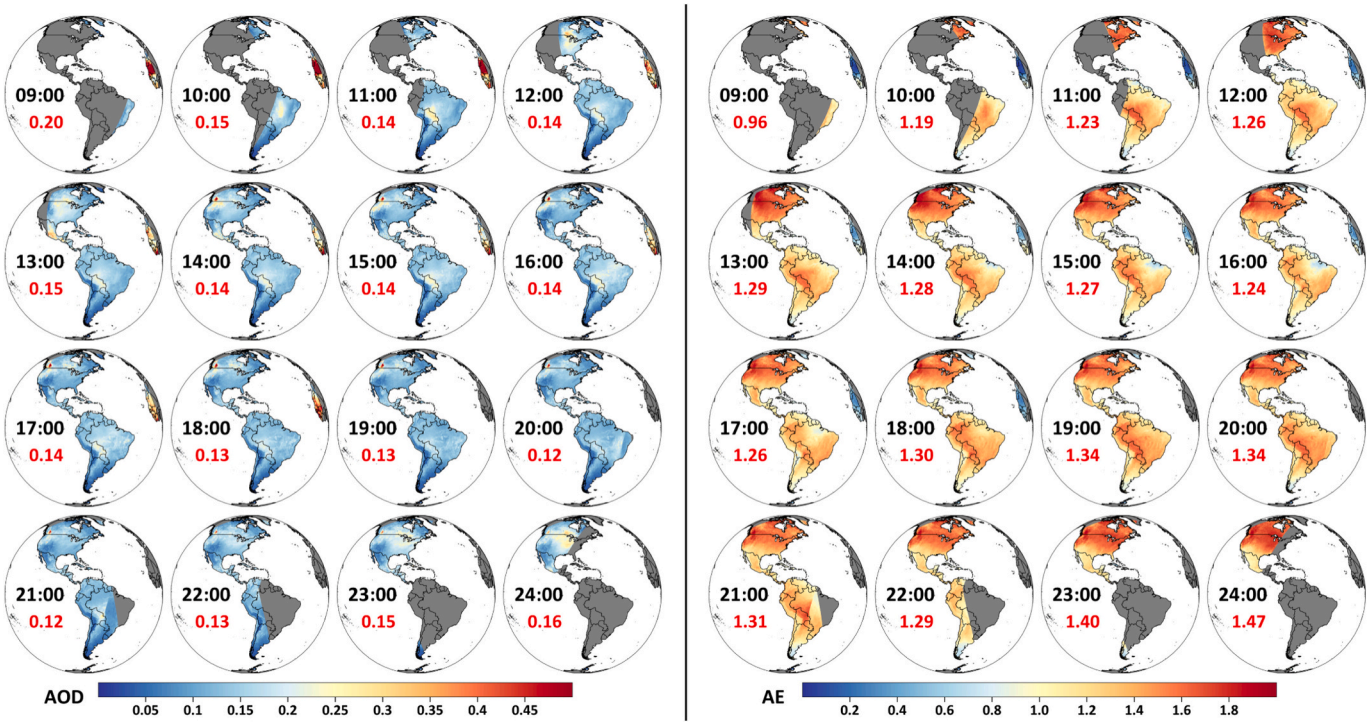

Fig. 11. Diurnal variations in the spatial distribution of AOD (550 nm) and Ångström exponent (440–1020 nm) from 09:00 to 24:00 UTC across the Americas in 2021, obtained via polynomial fitting of multi-band AOD retrievals (440, 500, 675, 870, and 1020 nm). Inset black and red numbers indicate the hours and mean values, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

composition and particle size. Desert regions are dominated by coarse-mode mineral dust, whereas biomass-burning regions are dominated by fine-mode absorbing aerosols. These contrasting optical characteristics result in different wavelength dependences of AF. Temporally, similar to SSA, AF exhibits minimal variability during daytime hours across the Americas, with domain-averaged AF varying by less than 0.03 throughout the day and a modest diurnal amplitude of less than 7% across wavelengths, indicating that aerosol scattering anisotropy remains stable throughout the day (Fig. 7m-p).

4.3. *Applications using multi-channel aerosol property retrievals*

This study highlights the advantages of multi-channel aerosol property retrievals for direct applications. These include improved retrieval of AOD at 550 nm and the corresponding derivation of the Ångström exponent (AE), enabling robust discrimination of aerosol size and composition. More importantly, the multi-parameter aerosol products enable continuous tracking of diurnal aerosol evolution during extreme events and facilitate the identification of source-related aerosol types.

4.3.1. *Enhanced characterization of aerosol size and type*

AOD at 550 nm and the Ångström exponent (AE, 440–1020 nm), derived from multi-wavelength AOD with improved accuracy compared to conventional approaches based on only one or two wavelengths, enable enhanced characterization of aerosol size and type (Fig. 11). The reconstructed 550-nm AOD shows spatial patterns consistent with other wavelengths, with persistently high values in localized regions. It also exhibits a clear diurnal cycle similar to that at other wavelengths, with elevated values at sunrise and sunset and a pronounced daytime evolution, reaching an amplitude of up to 55%. In contrast, AE exhibits a distinct diurnal pattern, increasing from morning to midday and subsequently decreasing toward afternoon, reflecting dynamic changes in aerosol size distributions. When interpreted together with AOD, these patterns provide insight into aerosol type and source characteristics. Regions with both high AOD and high AE (>1.6), indicative of fine-mode dominance, are primarily found over northeastern North America (e.g., the Great Lakes region), the Mexico–U.S. border region, and central South America. These regions show enhanced AE during daytime, likely driven by photochemical processes and biomass-burning emissions, and exhibit stronger spectral dependence. In contrast, regions with relatively high AOD but low AE (<0.6), mainly observed over northwestern Africa, northern South America, the Andes region, the southern tip of South America, and coastal areas, indicate coarse-mode aerosol dominance (e.g., dust and sea salt), often associated with relatively lower aerosol loading, resulting in weaker spectral dependence and reduced inter-band variability.

Fig. S14 illustrates the spatial distributions of scattering angle throughout the day for March, June, September, and December, indicating the geographic and seasonal variations in scattering geometry and potential hotspot biases in the satellite retrievals. Across seasons, scattering angles generally reach their lowest values between 16:00 and 18:00 UTC, particularly in March and September, indicating relatively lower aerosol sensitivity during this period. In contrast, other daytime hours exhibit more favorable scattering geometries for aerosol retrieval.

4.3.2. *Enhanced capture of aerosol evolution during wildfire events*

The multi-wavelength aerosol retrieval framework enables the derivation of multiple optical properties and composite indices, providing enhanced capability to characterize aerosol composition and identify extreme pollution events such as wildfires. During a severe wildfire event over the western United States on 30 August 2021 (Fig. 12), dense smoke plumes are evident in the false-colour satellite imagery. Official AOD retrievals exhibit substantial spatial gaps, particularly over and around dense smoke plumes, where traditional retrieval algorithms are often unable to provide valid retrievals. In contrast, our retrievals provide more spatially complete coverage and capture the overall spatial extent and diurnal evolution of aerosol loading, showing progressive intensification of smoke plumes from 15:00 to 21:00 UTC, followed by a gradual decline. However, AOD alone tends to underestimate the full spatial extent of wildfire impacts,

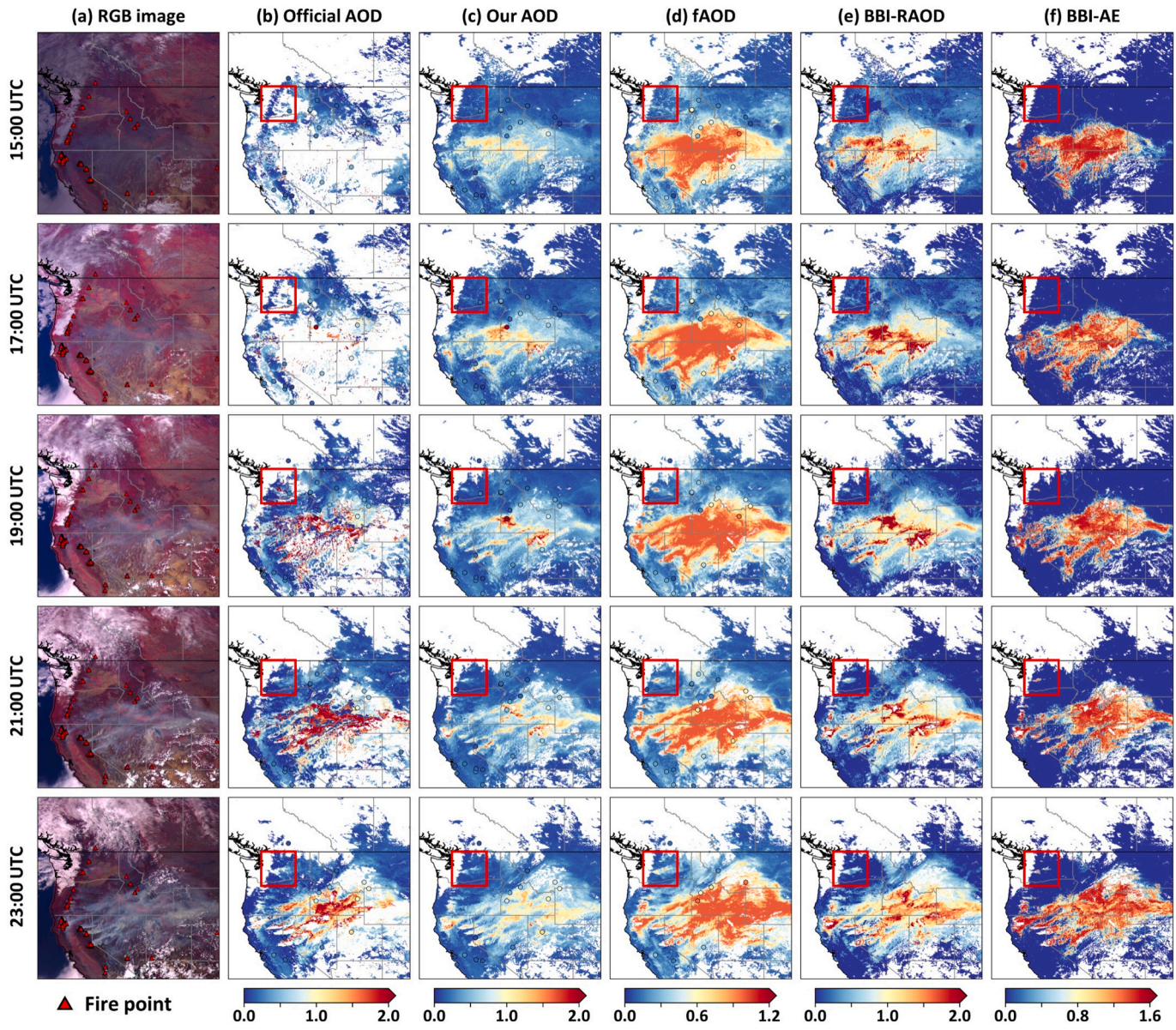

Fig. 12. Diurnal spatial patterns of (a) false-colour imagery (RGB: Bands 860–640–470 nm), (b) official AOD (550 nm), (c) our AOD (550 nm), (d) fine-mode AOD (fAOD; 500 nm), (e) biomass burning index based on the ratio of AOD (BBI-RAOD), and (f) biomass burning index based on AE (BBI-AE) over the western United States during a severe wildfire event on 30 August 2021. Red triangles in panel (a) indicate active fire locations, while colored dots in panels (b–d) indicate AERONET measurements. Fire data were obtained from the MCD14A1 active fire product. Red rectangles in panels (b-f) highlight relatively small fire events in Washington State. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

especially in downwind plume regions where transported smoke is not adequately represented. In contrast, fine-mode AOD (fAOD) more effectively delineates fire-related aerosols, revealing stronger and more spatially extensive smoke plumes than AOD alone. This highlights the dominant contribution of fine-mode particles from biomass-burning emissions and improves the representation of smoke evolution.

Building on this, the biomass burning index (BBI) provides even stronger discrimination of fire-related aerosols. The RAOD-based BBI enhances the identification of both active fire locations and transported smoke by incorporating aerosol absorption properties. Furthermore, the AE-based BBI exhibits sharper contrast and more precise localization, with high values tightly collocated with fire regions and near-zero values elsewhere, offering the most distinct separation between biomass-burning aerosols and background conditions. Significantly improved detection of the smoke plume, with even relatively small fire events in Washington State being successfully captured (as highlighted by the red rectangles). A similar wildfire event over the eastern United States on 7 September 2021 further demonstrates the robustness of this framework (Fig. S15). These results demonstrate that integrating multi-parameter aerosol properties provides a more comprehensive and physically interpretable framework for detecting and tracking wildfire smoke than AOD alone.

4.4. Comparison with official aerosol product

Finally, we demonstrate the enhanced performance of the AeroCat-Trans model relative to the operational aerosol product (Fig. 13). During a wildfire case (3 September 2021), the official product exhibits fragmented and discontinuous spatial patterns, characterized by unrealistically high AOD values near active fire sources and abnormally low values in surrounding regions. In addition, it contains extensive missing data, particularly in fire-affected areas with heavy smoke. In contrast,

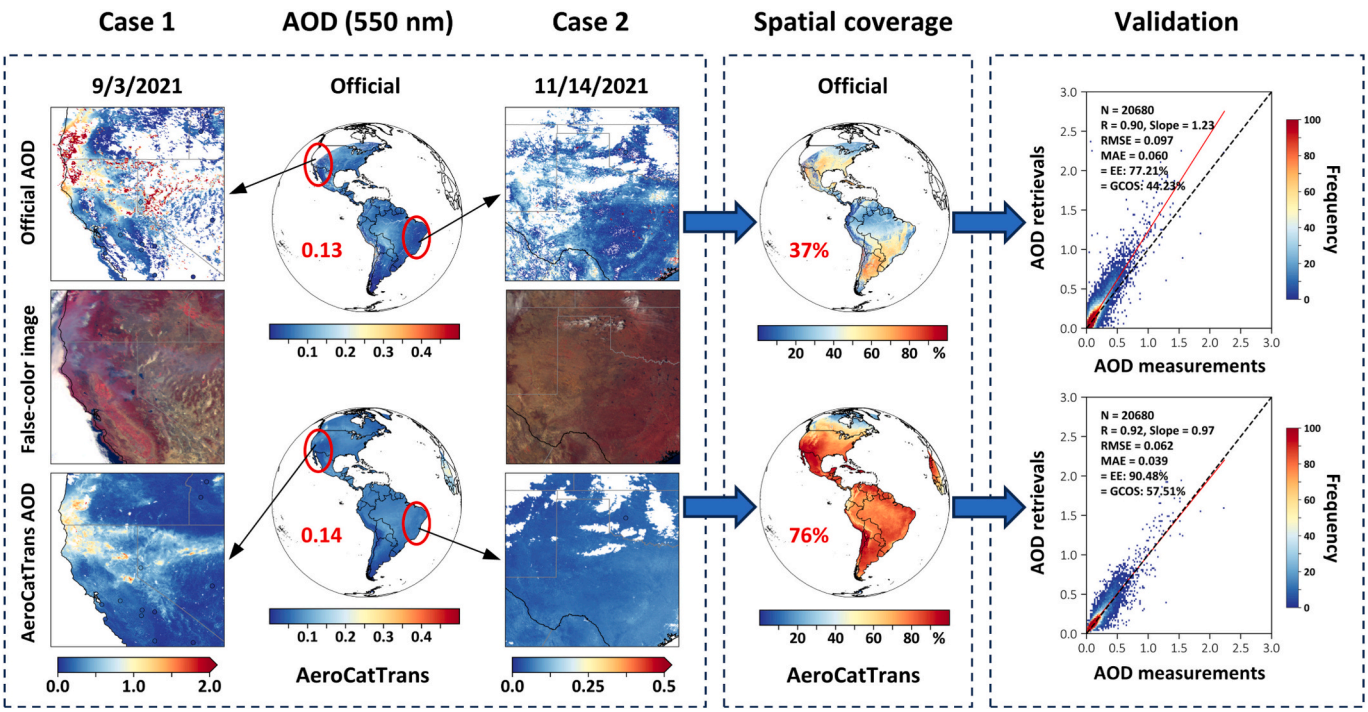

Fig. 13. Comparison of the official AOD product and the AeroCatTrans retrieval in terms of spatial distribution (left), spatial coverage (middle), and validation against AERONET AOD measurements (right). Two representative cases (3 September 2021 and 14 November 2021) are shown to illustrate differences in spatial completeness. The dashed black lines represent the 1:1 reference lines, and the red lines represent the fitted linear regression lines. Inset red numbers indicate the spatially averaged AOD and coverage. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

AeroCatTrans produces spatially continuous AOD fields, more realistically capturing both the intensity and spatial extent of smoke plumes. Under a clean-sky case (14 November 2021), the official product shows substantial retrieval gaps over bright surfaces, mainly due to limitations of the dark-target algorithm. In contrast, AeroCatTrans maintains stable and continuous retrievals, significantly improving spatial consistency. These improvements are further reflected in spatial coverage, with AeroCatTrans increasing mean coverage from 37% to 76%, and the most pronounced gains occurring over North America, especially in the western United States, where the official product frequently suffers from retrieval gaps.

In addition to improved spatial completeness, AeroCatTrans demonstrates substantially enhanced retrieval accuracy. We used the highest-quality ABI AOD retrievals with sufficient valid coverage (>20%) within an approximately 10 × 10-pixel window for comparison. Pixel-to-pixel collocation was performed, and only paired samples with valid retrievals from both products were retained. Although the official product shows a similar annual mean AOD of 0.13 ± 0.09, compared with 0.14 ± 0.04 for our retrievals, our framework shows improved overall accuracy relative to the official ABI AOD product (e.g., spatial CV-$R = 0.92$ vs. 0.90, RMSE = 0.062 vs. 0.097, and the percentage within EE = 90.48% vs. 77.21%; Fig. 13). However, our retrievals exhibit smoother spatial patterns and tend to underestimate AOD under high aerosol loading conditions. Moreover, the official products tend to overestimate AOD compared with our retrievals (Slope = 1.23 vs. 0.97), which may be partly attributed to residual cloud contamination caused by imperfect cloud masking. This issue is particularly evident over regions with frequent cloud occurrence, such as tropical areas.

5. Summary and conclusions

Unlike traditional machine-learning approaches that typically retrieve only a limited set of aerosol parameters (e.g., AOD), this study develops a novel multi-parameter, multi-wavelength joint-retrieval framework for aerosol optical properties (AeroCatTrans) by integrating CatBoost and a Transformer to leverage their complementary strengths. Our framework enables the joint retrieval of five key hourly aerosol optical properties from GOES-16/ABI imagery, including multi-wavelength aerosol optical depth (AOD at 440, 500, 675, 870, and 1020 nm), fine-mode AOD (fAOD at 500 nm), coarse-mode AOD (cAOD at 500 nm), single-scattering albedo (SSA at 440, 675, 870, and 1020 nm), and the asymmetry factor (AF at 440, 675, 870, and 1020 nm).

AeroCatTrans demonstrates consistently high and stable accuracy for hourly retrievals of aerosol optical properties throughout the day, particularly for multi-band AOD. Sample-based CV-R (RMSE) values reach 0.92 (0.110), 0.92 (0.094), 0.91 (0.060), 0.90 (0.042), and 0.90 (0.035) at 440, 500, 675, 870, and 1020 nm, respectively. The model also performs robustly in retrieving fine- and coarse-mode AOD at 500 nm, with CV-R (RMSE) values of 0.93 (0.082) and 0.89 (0.026), respectively. For more challenging parameters, AeroCatTrans achieves CV-R values ranging from 0.53 to 0.72 for SSA and from 0.88 to 0.93 for AF across wavelengths under sample-based CV. Under the stricter spatiotemporal-blocked CV framework, the model maintains strong performance in retrieving multi-band AOD ($R = 0.75$–0.81), fAOD ($R = 0.83$), and cAOD ($R = 0.79$), while CV-R values range from 0.32 to 0.56 for multi-band SSA and from 0.81 to 0.88 for AF. Nevertheless, limited observations in under-sampled regions may reduce the model's ability to learn region-specific aerosol characteristics, resulting in greater retrieval uncertainty. Moreover, AeroCatTrans substantially improves upon the GOES-16 operational product, enhancing spatial coverage by 105%, while increasing the reliable retrievals by 17% falling into the EE and reducing RMSE by 36% relative to ground-based observations.

Using the retrieved hourly aerosol properties, we reveal distinct diurnal behaviors across the Americas. Multi-band AOD exhibits substantial variability, generally decreasing throughout the day, with average amplitudes ranging from 53% to 56%. The diurnal amplitudes of fAOD and cAOD indicate moderate variability, with average amplitudes of 59% and 39%, respectively. In contrast, SSA and AF exhibit

much weaker diurnal variability, with amplitudes of 4%–6% and 4%–7%, respectively. These multi-channel aerosol retrievals further enhance the estimation of AOD at 550 nm and the derivation of the Ångström exponent (AE), enabling improved discrimination of aerosol size and composition. Moreover, additional indices derived from multiple aerosol parameters, such as the biomass burning index (BBI), facilitate the tracking of fine-mode smoke evolution during extreme wildfire events. Overall, these results highlight the critical role of multi-parameter and multi-wavelength aerosol retrievals in providing a more robust and physically interpretable characterization of aerosol types and emission sources than AOD alone. Nevertheless, the retrieval of SSA and AF from GOES/ABI imagery remains challenging. Further efforts are needed, including the incorporation of additional constraints, such as multi-angle or polarization observations, as well as more reliable ground-based measurements.

While the present study focuses on the development of the aerosol retrieval framework, future work will extend our approach to additional satellites in the GOES-R series (e.g., GOES-19) and to substantially longer observation periods. We will also consider multiple and larger AERONET-centered spatial windows to increase regional sampling diversity, together with appropriate quality-control procedures to reduce the potential influence of cloud contamination and spatially heterogeneous events on model training and evaluation.

CRediT authorship contribution statement

Yulong Fan: Writing – original draft, Validation, Software, Resources, Methodology, Formal analysis, Data curation. **Zhanqing Li:** Writing – review & editing. **Lin Sun:** Writing – review & editing. **Zhihui Wang:** Data curation. **Xiaohang Shi:** Data curation. **Fan Cheng:** Data curation. **Zhaoyu Wang:** Data curation. **Yu Qu:** Data curation. **Jing Wei:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2026.115719.

Data availability

The ground-based AERONET measurements are available at https://aeronet.gsfc.nasa.gov/. The GOES-16/ABI satellite observations and corresponding cloud masks are available at https://test.icare.univ-lille.fr/. The GOES-16 official aerosol product is available at https://noaa-goes16.s3.amazonaws.com/. The GEOS-CF data is available at https://portal.nccs.nasa.gov/datashare/gmao/geos-cf. The reanalysis MERRA-2 data is available at https://disc.gsfc.nasa.gov/. DEM is available at https://www.ncei.noaa.gov/products/. The MCD14A1 active fire product (Version 6.1) is available at https://ladsweb.modaps.eosdis.nasa.gov/.

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